

Skill Arena - Phase 5 Document 2

API Architecture, Service Communication, Event Bus Design & Real-Time Messaging Framework

Document Purpose

Define how services communicate, how APIs are structured and how real-time events move throughout the platform.

Business Objective

Enable secure, scalable and reliable communication between users, services and platform components.

API Architecture

Central API Gateway with versioned APIs, authentication controls and service routing.

API Design Principles

Consistency, security, scalability, observability and backward compatibility.

API Categories

Public APIs, Authenticated Player APIs, Administrative APIs and Internal Service APIs.

API Gateway Responsibilities

Authentication, authorization, rate limiting, routing, logging and monitoring.

REST API Layer

Primary interface for standard platform operations and data retrieval.

Real-Time Communication Layer

WebSockets and event streams for gameplay, matchmaking and live updates.

Service Communication Model

Combination of synchronous API calls and asynchronous event-driven messaging.

Event Bus Architecture

Central message bus used to distribute events between services.

Event Producers

Gameplay Service, Wallet Service, Tournament Service, User Service and Security Service.

Event Consumers

Notifications, Reporting, Analytics, Treasury, Compliance and Replay Services.

Core Event Types

User Events, Match Events, Reward Events, Financial Events, Security Events and Tournament Events.

Match Events

Match Created, Match Started, Match Completed, Replay Generated and Reward Issued.

Financial Events

Deposit Completed, Withdrawal Requested, Reward Issued and Ledger Updated.

Notification Events

Email Sent, Alert Generated, Tournament Reminder and Security Alert.

Replay Events

Replay Created, Replay Verified and Replay Archived.

Message Reliability

Guaranteed delivery, retry policies, dead-letter queues and event persistence.

Observability

Distributed tracing, event monitoring and service performance tracking.

Scalability Requirements

Support millions of users, high event throughput and global expansion.

Fault Tolerance

Event replay, message persistence and graceful service degradation.

Security Requirements

Encrypted communication, service authentication and strict API authorization.

Monitoring Requirements

API health monitoring, event bus monitoring and message delivery monitoring.

Audit Requirements

API requests, service interactions and event processing activities logged.

Performance Requirements

Low-latency communication for gameplay and real-time platform operations.

Acceptance Criteria

Services communicate reliably, securely and efficiently under large-scale workloads.